Intelligent Edge-Based Laboratory Access Monitoring System

Restricted labs require monitoring to prevent accidents and equipment damage.

Manual surveillance is labour-intensive and cannot maintain continuous attention.

Traditional passive observation delays response to safety incidents.

Proposed solution: Distributed edge-AI system for real-time detection and alerts.

Problem Statement & Objectives

Problem Statement

- Manual surveillance is slow and error-prone.
- Cloud monitoring adds latency, network dependency, and privacy concerns.
- Single-sensor or vision-only systems risk false detections.

Objectives

- Edge monitoring using Raspberry Pi Zero 2 W with webcam running MobileNet for real-time person detection, triggered by PRI sensor.
- Frames with detected persons sent via MQTT to central Raspberry Pi.
- Central Raspberry Pi performs YOLO26-nano detection for confirmed frames.
- Multi-source validation via PRI sensor
- Facial recognition for whitelist/blacklist to enforce access permission.
- Evaluate detection accuracy, latency, and system stability.

System Architecture Design

Distributed Edge-Fog Laboratory Monitoring Architecture with Web Interface

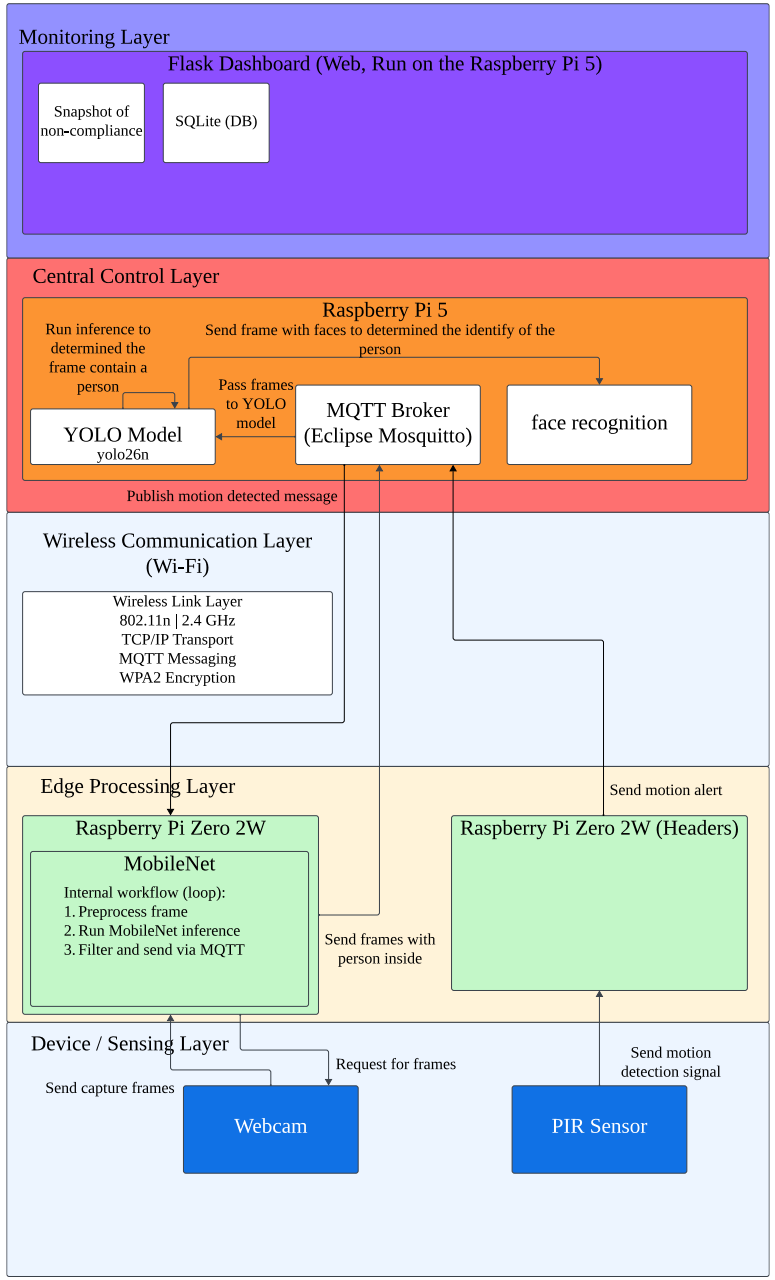

Figure 1: Five-layer architecture showing data flow from sensors to web monitoring

Design and Justification

Architecture	• Distributed edge and central inference • Event-driven processing
Model Strategy	• Motion-trigger activation • Lightweight edge filtering • High-accuracy central model
Efficiency	• Cropped transmission • Reduce bandwidth use
Communication	• Publish and Subscribe messaging

Feasibility, Impact & Risk

Feasibility

- Low-power edge nodes
- Off-the-shelf hardware
- Modular deployment

Impact

- Automated monitoring
- Faster violation detection
- Reduced manual supervision

Risks

- False motion triggers
- Lighting conditions
- Hardware limits
- Network disruption

Mitigation

- Vision verification
- Model optimisation
- Local event logging

YouTube Link

- https://youtu.be/4luxrJ1sUvs